

Memory-Centric Agentic Inference — cycles 19-21

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Memory-Centric Agentic Inference — cycles 19-21

Abstract

Cycles 19-21 moved the memory-centric architecture package from synthetic architecture and enforcement models toward deployment-facing measurement boundaries. The main question for these cycles was not whether Option B or Option C can be represented. Earlier milestones had already shown that:

- **Option A** is conventional request/model/KV-centric serving.
- **Option B** is a memory-object-aware runtime for reusable retrieved context, semantic cache entries, prefix state, and tool outputs.
- **Option C** is a trajectory/DAG-aware memory fabric for branch, verifier, durable workspace, and multi-agent state.

The new question was whether the remaining energy and contention claims can be calibrated without overclaiming. Cycle 19 built M-DC12-1, a host-local proxy harness for DC-001 and DC-002. DC-001 is the per-tier energy or dollar cost of bytes moved or retained. DC-002 is the p50/p95/p99 latency of CXL or pooled-memory tiers under contention. The proxy harness exercised the threshold plumbing, but every output remained labeled `host_local_proxy` and `production_calibrated=false`.

Cycle 20 built M-PRODTELEM-1, a production-target telemetry contract and ingestion harness. It defines the minimum schema future production rows must satisfy before they can update energy, economics, or CXL/contention claims. The auditor found one moderate issue: invalid rows previously retained positive granted credit. That was fixed. After the patch, all invalid or blocked rows grant zero reuse and energy credit, while valid synthetic production-shaped rows remain candidate-only and still do not calibrate production claims.

Cycle 21 supplied researcher, worker, and auditor session IDs, but no separate cycle-21 milestone, artifact, ledger entry, script, or data product was found in the workspace. This report treats cycle 21 as a reporting and consolidation boundary, not as a new technical milestone.

The package is now contract-ready for production measurement, but not production-calibrated. The strongest remaining boundary is empirical: real accelerator/host power counters, tier-specific bytes, target-topology CXL or pooled-memory latency tails, tenant concurrency, workload/object labels, and security decisions must be collected together before CL-012 or production Option B/C claims can be upgraded.

Introduction

The research program investigates whether future AI infrastructure should be designed around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. Earlier cycles built the workload taxonomy, lifetime model, cost model, simulator, scheduling comparison, architecture proposal, trace schema, queueing model, compression model, runtime prototype, calibration map, security model, synthesis matrix, measurement plans, energy/contention falsification harness, and security enforcement replay.

Cycles 19-21 focused on the empirical boundary left open by those artifacts. Public hardware and interconnect references establish that HBM, GPU memory, PCIe, CXL, NVMe, CPU memory, prefix caching, and semantic caching are relevant infrastructure layers [1]-[14]. The remaining question is whether agentic memory placement and reuse reduce measured energy, cost, latency, or capacity pressure on target systems.

The prior cycles had already narrowed the thesis:

`MemoryCentricArchitecture` is strongest when `RetainedStateValue > CoordinationCost + ValidationOverhead + RecoveryCost + ExpectedSecurityLoss + DurableTailCost + EnergyDollarContentionCost`.

Cycles 19-21 did not remove that condition. They made the measurement boundary stricter.

Methodology

The report consolidates workspace artifacts, plan and ledger records, the supplied audit report, figures, CSV row counts, and source session IDs. No callable `search_sessions` or `list_session_catalog` tool was available in this reporting environment. The supplied session IDs are therefore used as traceability anchors, while technical content is grounded in the files and audit records present in the workspace.

The chronological source map is:

Cycle	Session IDs	Main source material	Role in timeline
Pre-cycle context	M-PLAN-1 ledger events from cycle 18	<code>constrained_memory_planning.md</code> , <code>memory_plan_*.csv</code> , planner figures	Converts prior option-level architecture into per-object actions and constraints.
Cycle 19	researcher 690d5614-888c-48d0-aaed-a66968b3ba66; worker de2142ef-36dd-4408-a8fa-ce6cbc228f5a; auditor 29cc862f-f392-4fd7-95c0-364d7fae250b	<code>dc12_local_proxy_calibration.md</code> , <code>local_dc12_proxy_bench.py</code> , <code>apply_dc12_proxy_calibration.py</code> , <code>verify_dc12_proxy_calibration.py</code> , <code>dc12_*.csv</code> , <code>dc12_*.png</code>	Builds and validates host-local proxy plumbing for DC-001/DC-002.

Cycle	Session IDs	Main source material	Role in timeline
Cycle 20	researcher c623028a-c638-4bc9-9903-d800b74a2a9b; worker 4a0a5a95-7746-49dd-8a33-cf156aec49ec; auditor ba85f69f-1cd9-4558-9090-70e67092b679	production_dc12_ telemetry.md, build_production_ dc12_fixtures.py, ingest_production_ dc12_telemetry.py, verify_production_ dc12_telemetry.py, production_dc12_*.csv, production_dc12_*.png, supplied audit report	Defines production telemetry schema and fail-closed ingestion semantics.
Cycle 21	researcher 5fc0c156-46bb-46f6-b1ce-daa53a7b5f4f; worker 89b50d43-c27a-4f1b-9ebc-45b450af79c5; auditor d852e09d-a835-4d89-9fce-8ae079cbeafc	No separate workspace milestone or artifact found	Treated as a report- ing/consolidation boundary.

Results

Planning Context Before Cycle 19

Immediately before cycle 19, M-PLAN-1 converted the architecture from option labels into per-object planning actions. The planner consumes trace-v3 security decisions, runtime policy rows, compression safety scores, queueing reversal thresholds, CXL-contention thresholds, and energy/contention option collapses.

The planner expression was:

$$\text{NetPlanValue} = \text{SafeReuseValue} - \text{MovementCost} - \text{ValidationOverhead} - \text{QueueOverhead} - \text{ContentionPenalty} - \text{CompressionRisk}.$$

Security and compression safety are hard gates. Denied reuse forces `recompute_or_drop` with zero positive reuse credit. Unsafe compression cannot be selected. Capacity, queueing, validation overhead, and contention are soft constraints that can downgrade Option C to B/A, offload cold state, preserve pointers, or emit an infeasible row.

The auditor-validated planner emitted:

Output	Count
memory_plan_actions.csv	85 action rows
Unique objects represented	62
Workload summaries	6
Infeasible/downgrade rows	82
Hook ablations	12
Sensitivity rows	24

At workload level, RAG stayed Option B, both controls stayed Option A, code-agent loop and multi-agent branch/merge downgraded from Option C to Option B, and verification-heavy stayed Option C.



Cycle 19: Local DC-001/DC-002 Proxy Calibration

The harness measured:

- 4

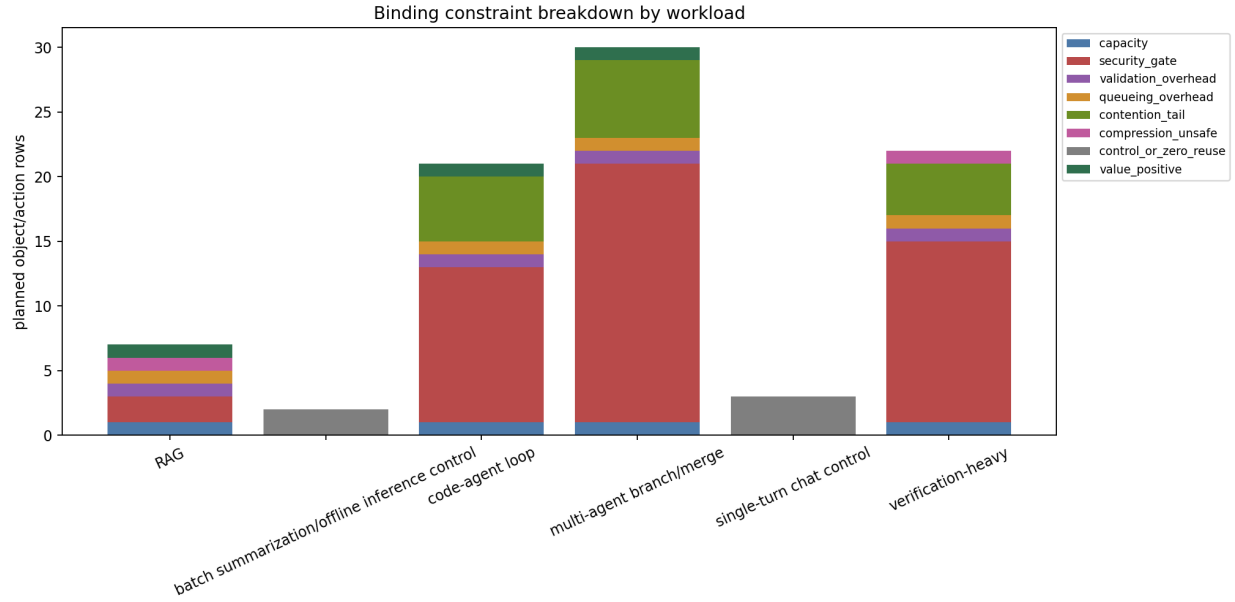


Figure 2: Binding constraint breakdown by workload, measured as planner row counts.

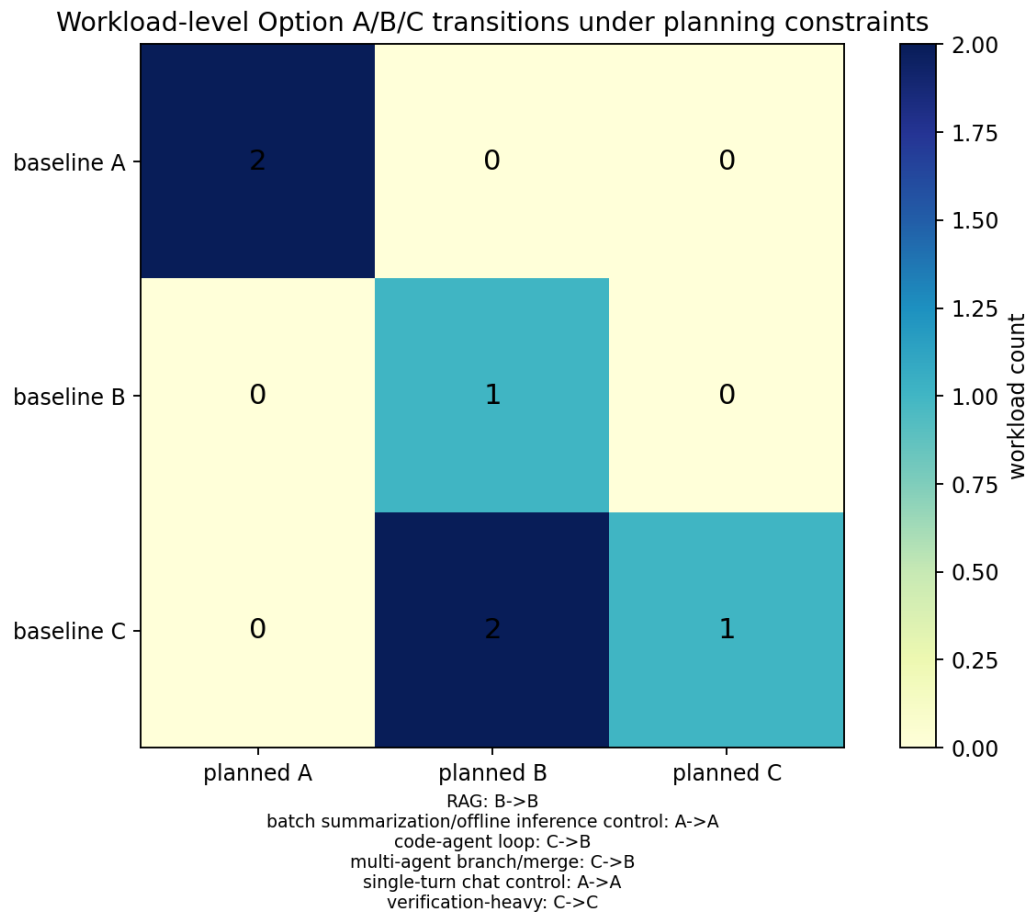


Figure 3: Workload-level Option A/B/C transitions before and after constrained planning.

The auditor found and fixed a moderate measurement-validity issue in the cycle 19 work. The original contention timing used Python threads, which risked measuring interpreter scheduling rather than memory contention. The fix replaced this with process-level shared-memory workers. After regeneration, the auditor validated the milestone.

Cycle 19 emitted:

Output	Rows
dc12_local_bench_metadata.csv	11
dc12_byte_movement_measurements.csv	12
dc12_contention_measurements.csv	3
dc12_proxy_threshold_overlay.csv	60
dc12_claim_update_matrix.csv	4
dc12_missing_production_telemetry.csv	5

The measured proxy values were intentionally reported with limits. Sequential copy throughput ranged from 4,343.735 MiB/s to 17,896.271 MiB/s. Sequential write throughput ranged from 344.407 MiB/s to 409.685 MiB/s. Random single-byte throughput ranged from 3.562 MiB/s at 256 KiB to 2.134 MiB/s at 4 MiB.

For contention, local p99 latency rose from 7.206 microseconds at one worker to 8.317 microseconds at four workers. The p99 contention proxy was 1.1542 times the one-worker baseline. That did not cross any non-control retained-value reversal threshold in `cxl_contention_thresholds.csv`.

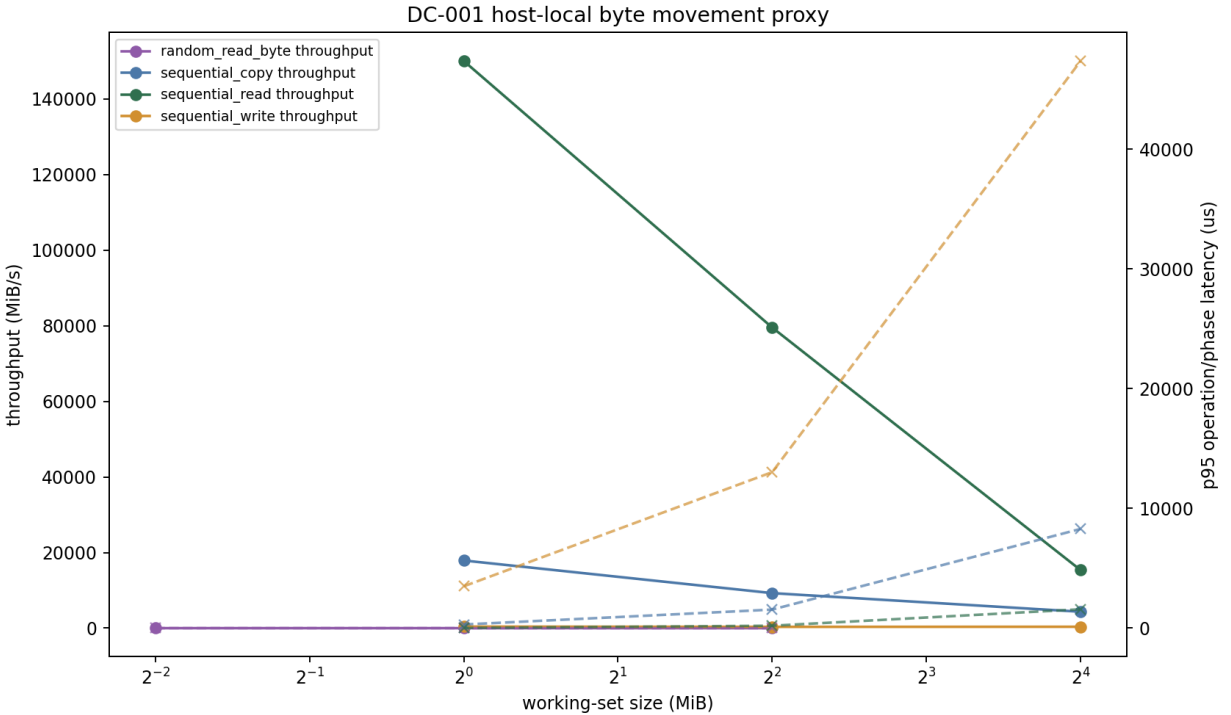


Figure 4: Measured host-local byte-movement throughput/latency proxy by working-set size and access pattern.

The claim update matrix preserved the boundary:

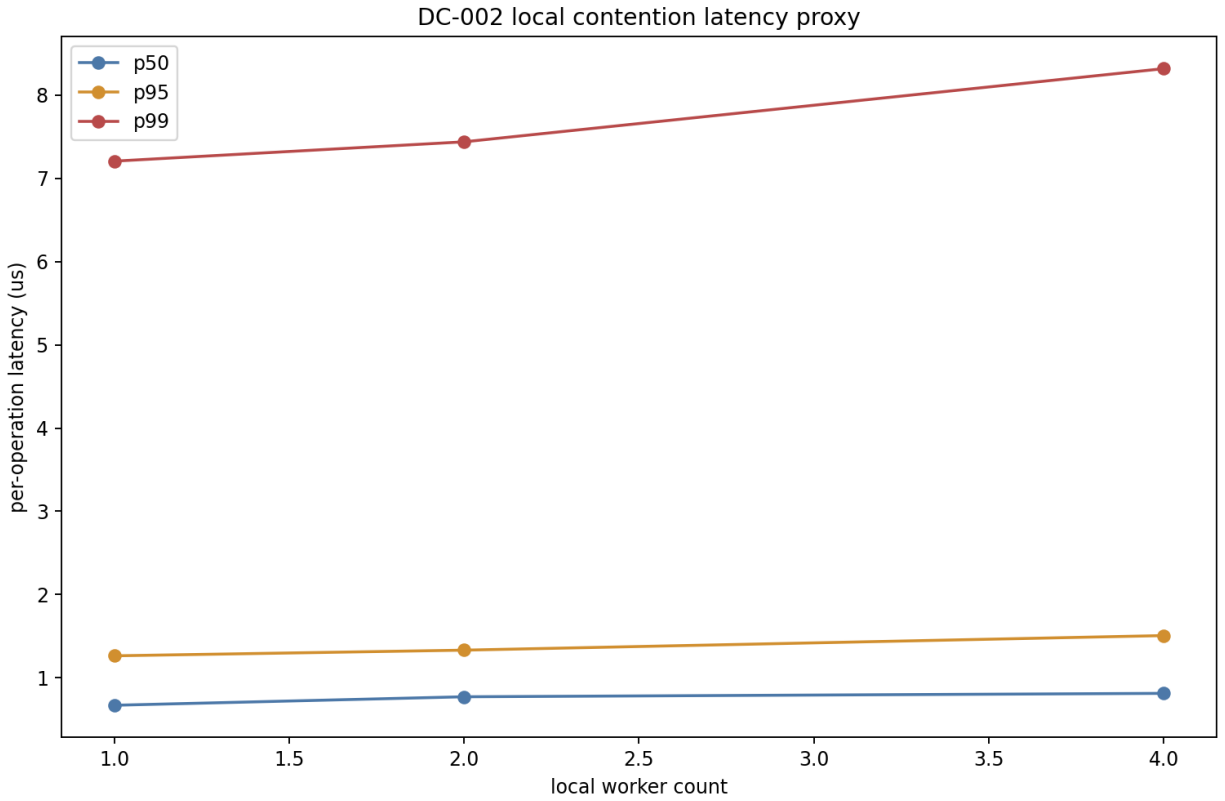


Figure 5: Measured latency p50/p95/p99 under increasing local contention.

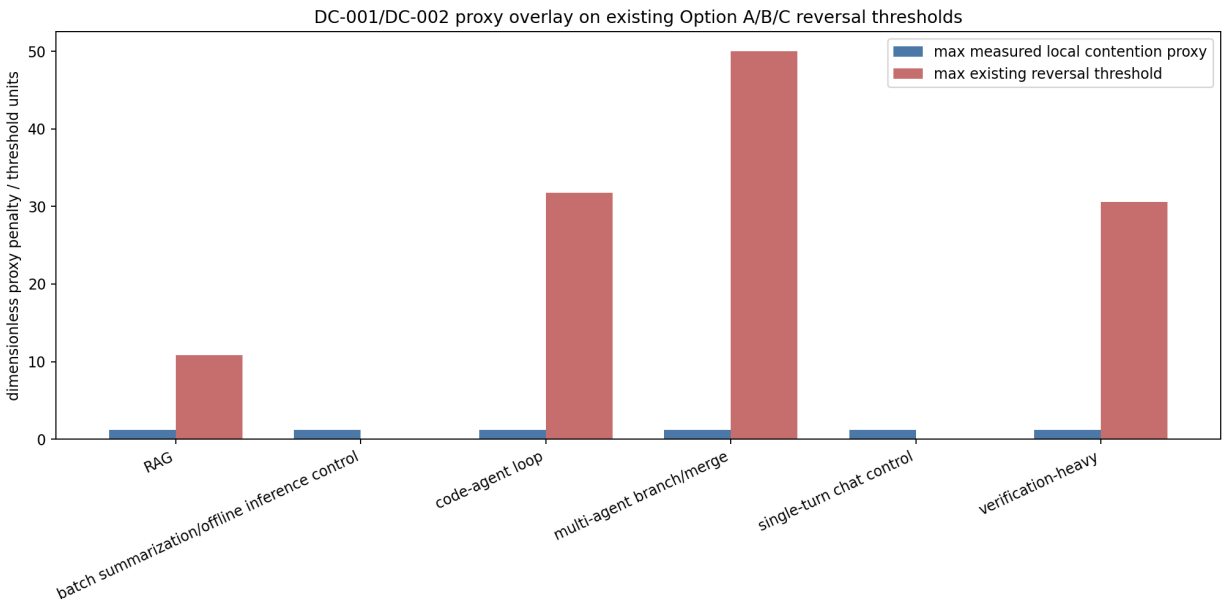


Figure 6: Proxy-measured DC-001/DC-002 regimes overlaid on existing Option A/B/C reversal thresholds.

Claim	Update status	Production calibrated?	Result
CL-012	proxy_only	false	Direct power counters unavailable; energy/economics remain uncalibrated.
CL-004	proxy_threshold_overlay	false	No DC-002 rows crossed under local proxy contention; controls remained Option A.
CL-005	proxy_threshold_overlay	false	Option B/C updates require retained-value threshold crossings; host-local proxy is not CXL production evidence.
SECURITY-GATE-ENERGY-001	validated_negative_control	false	75 denied-reuse rows had zero safe reuse credit.

The main cycle 19 result is therefore a null result with engineering value. The system can ingest local proxy measurements and compare them to DC-001/DC-002 thresholds, but it does not promote proxy evidence into production calibration.

Cycle 20: Production DC-001/DC-002 Telemetry Contract

Cycle 20 added M-PRDTELEM-1, a production-target telemetry contract and ingestion harness. The research decision was to turn the missing production evidence categories from cycle 19 into an executable schema and fixture suite.

The production acceptance rule is fail-closed:

```
ProductionCalibrated = evidence_label=production_target AND required fields present AND
join keys valid AND power/byte intervals aligned AND delta above noise AND security/provenance/retention/verifier gates pass AND threshold is comparable.
```

The schema covers 39 required rows of telemetry fields. The fields include production target identity, hardware topology, accelerator type, source and destination tier, workload and object class, architecture option, reuse decision, bytes moved, resident bytes, interval timing, measured joules, power-counter source, noise floor, CXL or pooled-memory p50/p95/p99 latency, tenant count, queue depth, security/provenance/retention/verifier gates, and interval alignment.

Cycle 20 emitted:

Output	Rows
production_dc12_telemetry_schema.csv	39

Output	Rows
production_dc12_valid_fixture.csv	4
production_dc12_invalid_fixtures.csv	8
production_dc12_ingestion_results.csv	12
production_dc12_threshold_replay.csv	12
production_dc12_claim_update_matrix.csv	5
production_dc12_missing_fields_report.csv	5

The valid fixtures are complete synthetic production-shaped rows. They prove the ingestion path but use `evidence_label=synthetic_production_fixture`, so they remain `production_calibrated=false`. Three rows became candidate-only rows. The control fixture stayed non-candidate with zero reuse and energy credit.

The invalid fixtures cover missing power intervals, missing tier-specific bytes, missing workload/object labels, missing topology or tenant concurrency, unaligned power and byte intervals, below-noise DC-001 energy, security-denied positive credit, and a host-local proxy row mislabeled for production ingestion.

The supplied audit report records one moderate issue. `production_dc12_ingestion_results.csv` previously granted positive reuse or energy credit to invalid blocked rows. The fix changed `scripts/ingest_production_dc12_telemetry.py` so granted credit requires no `blocked_reason` and passing security/provenance gates. The verifier now asserts every invalid fixture has zero granted reuse and energy credit. Documentation in `production_dc12_telemetry.md` was updated to match.

After the patch:

Ingestion category	Count
Ingestion rows	12
Calibration candidates	3
Production-calibrated rows	0
Invalid rows with named <code>blocked_reason</code>	8
Invalid rows with granted reuse/energy credit	0

Blocked reasons were:

Blocked reason	Rows
<code>missing_required_field</code>	3
<code>power_byte_interval_mismatch</code>	2
<code>below_noise_floor</code>	1
<code>security_denied_positive_credit</code>	1
<code>not_production_evidence_label</code>	1

The production claim matrix is now contract-ready but not calibrated:

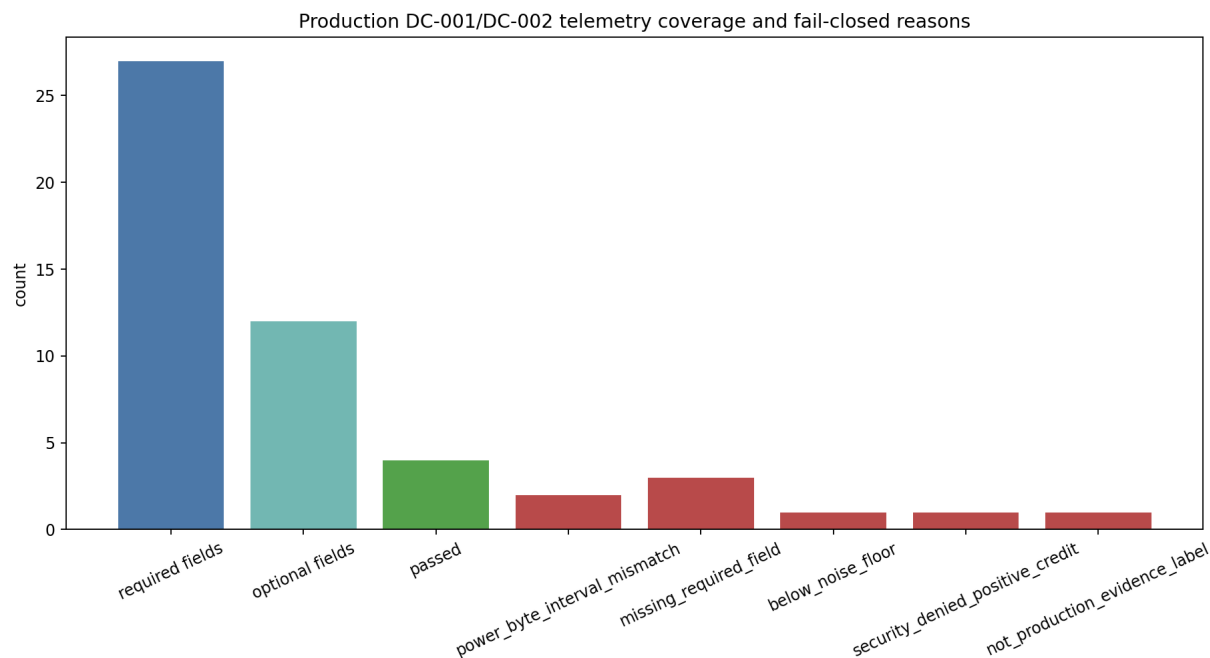


Figure 7: Production telemetry schema coverage and fail-closed reasons by fixture class.

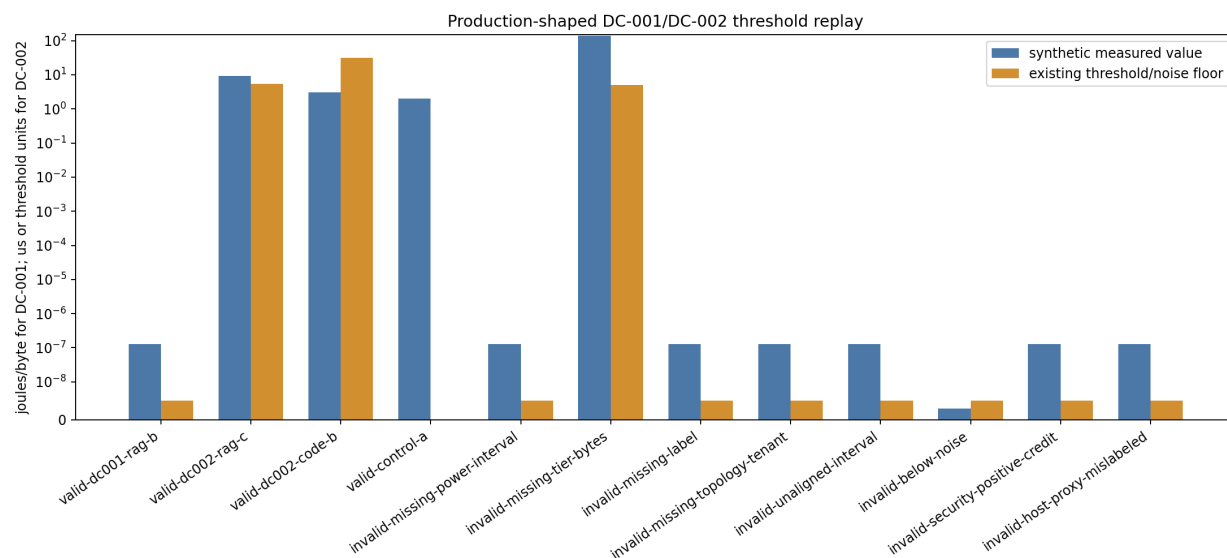


Figure 8: Synthetic production-shaped DC-001/DC-002 measurements overlaid on existing noise and contention thresholds.

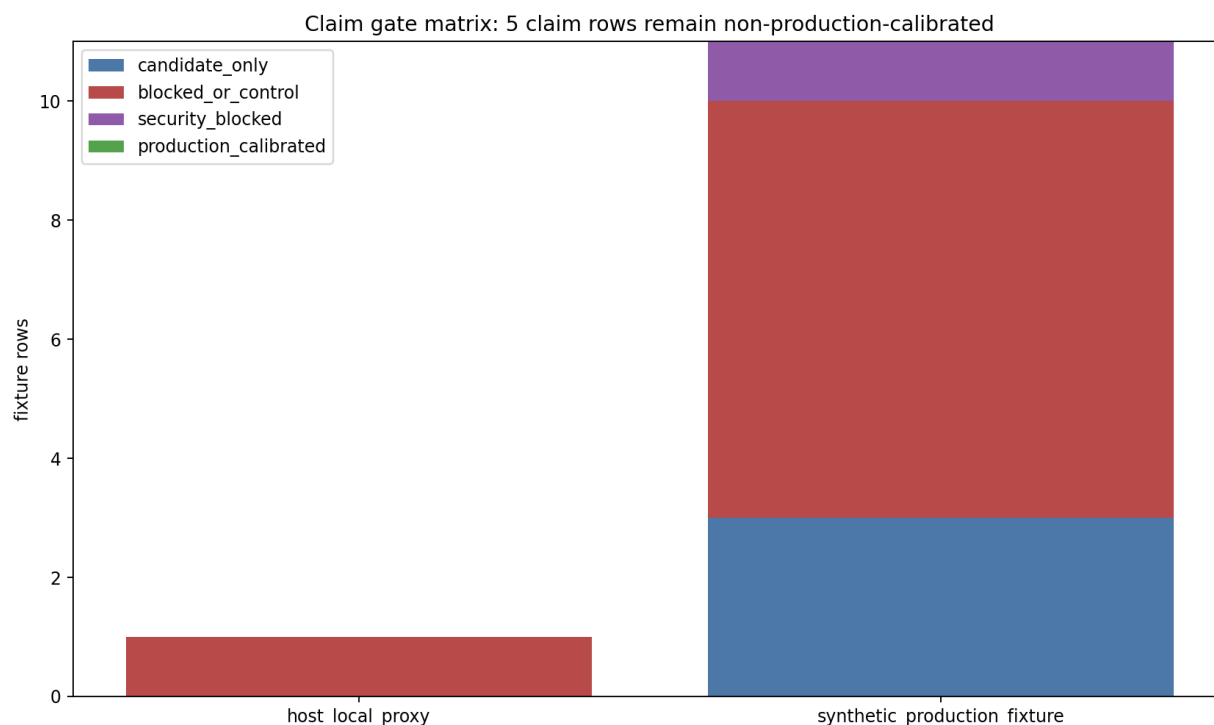


Figure 9: Claim gate matrix by evidence class, candidate status, security gate, and production-calibration status.

Claim	Status	Production calibrated?	Boundary
CL-012	production_contract_ready_candidate_only	false	Three synthetic production-shaped rows passed gates, but no real production_target evidence was ingested.
CL-004	threshold_replay_ready	false	DC-002 rows can replay against existing CXL/pooled-memory thresholds.
CL-005	contention_gate_ready	false	Option B/C updates are named by threshold crossings only.
SECURITY-GATE-ENERGY-001	validated_negative_control	false	Security-blocked rows grant zero credit.

Claim	Status	Production calibrated?	Boundary
CONTROL-OPTION-A	validated_negative_control	false	Control rows remain Option A and are not calibration candidates.

The cycle 20 result is an enforceable production evidence boundary. Future telemetry can update claims only if it is direct target evidence and passes the schema, interval, noise, security, provenance, retention, verifier, and comparability gates.

Cycle 21: Reporting Boundary and Record Gap

Cycle 21 supplied researcher, worker, and auditor session IDs, but no separate cycle-21 artifact was found in the workspace. Searches across `promise_ledger.jsonl`, `plan_of_record.md`, `memory-centric-agentic/`, `scripts/`, `tests/`, `data/`, and `reports/cycles/` found no cycle-21 milestone, markdown file, script, test, CSV, figure, or ledger event.

This is not reported as a technical defect in the architecture package. It is a source-record gap for the reporter. The technical report for cycles 19-21 therefore covers the validated cycle 19 and cycle 20 work and treats cycle 21 as consolidation/reporting.

Discussion

Cycles 19-21 changed the status of the package in a specific way. They did not prove production energy savings or production CXL/pool-memory latency benefits. They made those claims mechanically testable without weakening the evidence boundary.

Before these cycles, energy/economics and CXL/contention were represented by synthetic sensitivity thresholds. After cycle 19, host-local proxy measurements can flow through the same threshold machinery. After cycle 20, future production rows have a schema and ingestion gate strict enough to prevent synthetic fixtures, host-local proxies, missing fields, below-noise deltas, interval mismatches, or security-denied reuse from updating production claims.

The resulting deployment-readiness levels are:

Layer	Status after cycles 19-21
Architecture options A/B/C	Mechanism validated in synthetic and replay artifacts; not a blanket production recommendation.
Per-object planner	Executable synthetic planning pass with security, capacity, queueing, contention, validation, and compression constraints.
Host-local DC-001/DC-002 proxy	Plumbing validated; external validity explicitly limited.
Production telemetry ingestion	Contract-ready with fail-closed fixtures and claim gates.
Production energy/economics claim CL-012	Still blocked.
Production Option B/C recommendation	Still conditional on real target telemetry and safe reuse evidence.

The architecture recommendation remains conservative:

- Use Option A for controls and workloads without positive safe non-KV retained value.
- Use Option B for object-local reuse only when provenance, freshness, tenant/cache isolation, invalidation, and security gates pass.
- Use Option C only when branch, verifier, trajectory, or durable workspace value survives security, queueing, compression, capacity, validation, contention, energy, and dollar costs.

The most important rule preserved across these cycles is zero unsafe credit. Reuse that is missing required fields, below noise, interval-mismatched, host-local only, security-denied, or provenance-invalid cannot contribute positive retained value, energy credit, or production calibration.

Conclusions and Recommendations

Cycles 19-21 make the memory-centric package more deployment-ready but not production-proven.

First, M-DC12-1 shows that DC-001/DC-002 threshold plumbing can ingest measured rows and preserve external-validity labels. Its host-local proxy results did not cross non-control retained-value reversal thresholds, and CL-012 stayed proxy_only.

Second, M-PRODTELEM-1 defines the production telemetry contract needed to upgrade DC-001/DC-002 from synthetic or proxy evidence to production evidence. The auditor patch closed the main accounting issue: blocked invalid rows now grant zero reuse and energy credit.

Third, the next work should collect real production_target telemetry with all required joins present at once: accelerator/host power counters, tier-specific bytes, target CXL or pooled-memory p50/p95/p99 latency, tenant concurrency, queue depth, workload/object labels, reuse decisions, and security/provenance/retention/verifier outcomes.

The immediate research agenda is therefore:

1. Run the production telemetry contract on target accelerator/host systems.
2. Join power, byte movement, latency, workload, object, and security decisions into single comparable rows.
3. Keep CL-012 and production Option B/C claims blocked until production_calibrated=true rows exist.
4. Preserve fail-closed accounting: invalid rows, host-local proxies, below-noise rows, and security-denied reuse receive zero granted credit.
5. Use the planner and production ingestion outputs together to identify which workload/object classes should be tested first.

References

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[14] CXL Consortium, “CXL Consortium Releases Compute Express Link 2.0 Specification,” Business Wire, 2020. <https://www.businesswire.com/news/home/2020110005037/en/CXL-Consortium-Releases-Compute-Express-Link-2.0-Specification>

Appendix: Implementation Details

Code Organization

Cycle 19 added or updated the following M-DC12-1 files:

File	Purpose
scripts/local_dc12_proxy_bench.py	Host-local byte movement and process-level shared-memory contention proxy benchmark.
scripts/apply_dc12_proxy_calibration.py	Maps proxy rows onto existing DC-001/DC-002 thresholds and claim updates.
scripts/plot_dc12_proxy_calibration.py	Renders proxy byte-movement, contention, and threshold-overlay figures.
tests/verify_dc12_proxy_calibration.py	Verifies proxy labels, negative controls, overlays, claim boundaries, and figures.
memory-centric-agentic/dc12_local_proxy_calibration.md	Documents measurement design, results, threshold overlays, and external-validity limits.

Cycle 20 added or updated the following M-PRODTELEM-1 files:

File	Purpose
scripts/build_production_dc12_fixtures.py	Builds valid and invalid production-shaped telemetry fixtures.
scripts/ingest_production_dc12_telemetry.py	Validates schema, enforces fail-closed blocked rows, grants credit only to allowed rows, and replays thresholds.
scripts/plot_production_dc12_telemetry.py	Renders production telemetry coverage, threshold replay, and claim-gate figures.
tests/verify_production_dc12_telemetry.py	Verifies schema coverage, invalid fixture blocking, candidate-only valid rows, zero invalid credit, and figures.
memory-centric-agentic/production_dc12_telemetry.md	Documents the production telemetry contract, fixture semantics, claim updates, and open deployment fields.

The current workspace manifest was updated for cycles 19-21. It records 35 Python scripts in `scripts/`, 4 Wolfram scripts, 6 test scripts, 10,716 total scripts/ lines, 21 model/synthesis markdown files, 93 CSV data/model files, and 44 figures.

Figure Inventory

Figures referenced in this report:

Figure	Dimensions
data/memory_plan_action_mix.png	2380 x 1570
data/memory_plan_constraint_breakdown.png	2040 x 1020
data/memory_plan_option_transitions.png	1360 x 1020
data/dc12_byte_movement_proxy.png	1700 x 1020
data/dc12_contention_latency_proxy.png	1530 x 1020
data/dc12_threshold_overlay.png	2040 x 1020
data/production_dc12_telemetry_coverage.png	1870 x 1020
data/production_dc12_threshold_replay.png	2210 x 1020
data/production_dc12_claim_gate_matrix.png	1700 x 1020

Validation Results

The supplied audit report for M-PRODTELEM-1 reports:

- Critical issues: none.
- Moderate issues: one found and fixed.
- Fix: invalid blocked rows now grant zero reuse and energy credit.
- Post-fix schema rows: 39.
- Valid fixture rows: 4.
- Invalid fixture rows: 8.
- Ingestion rows: 12.
- Threshold replay rows: 12.
- Claim update rows: 5.

- Missing-field report rows: 5.
- Calibration candidates: 3 synthetic production-shaped rows.
- Production-calibrated rows: 0.
- Invalid rows with named `blocked_reason`: 8.
- Invalid rows with granted reuse/energy credit: 0.
- CL-012: still non-production-calibrated.
- Host-local proxy row: blocked.
- Below-noise row: blocked.
- Controls: remain Option A.
- Figures: all three production telemetry figures nonblank.
- Validation commands: `build`, `ingest`, `plot`, `verifier`, `py_compile`, independent CSV/figure probe, `promise_check`, and `org_check` passed.

The ledger records cycle 19 M-DC12-1 as auditor-validated after replacing threaded contention timing with process-level shared-memory workers. The ledger records cycle 20 M-PRODTELEM-1 as auditor-validated after tightening fail-closed credit accounting.

Source Session References

Role	Session ID
Cycle 19 researcher	690d5614-888c-48d0-aaed-a66968b3ba66
Cycle 19 worker	de2142ef-36dd-4408-a8fa-ce6cbc228f5a
Cycle 19 auditor	29cc862f-f392-4fd7-95c0-364d7fae250b
Cycle 20 researcher	c623028a-c638-4bc9-9903-d800b74a2a9b
Cycle 20 worker	4a0a5a95-7746-49dd-8a33-cf156aec49ec
Cycle 20 auditor	ba85f69f-1cd9-4558-9090-70e67092b679
Cycle 21 researcher	5fc0c156-46bb-46f6-b1ce-daa53a7b5f4f
Cycle 21 worker	89b50d43-c27a-4f1b-9ebc-45b450af79c5
Cycle 21 auditor	d852e09d-a835-4d89-9fce-8ae079cbeafc

Cross-Reference Map

Origin	Consuming artifact	Flow
<code>data/security_enforcement_decisions.csv</code>	<code>scripts/constrained_memory_planner.py</code>	Security-adjusted safe reuse and denied/downgraded rows become planner hard gates.
<code>data/cxl_contention_thresholds.csv</code>	<code>scripts/apply_dc12_proxy_calibration.py</code>	Existing DC-002 thresholds become comparison targets for host-local contention proxies.
<code>data/energy_measurement_requirements.csv</code>	<code>scripts/local_dc12_proxy_bench.py</code> and <code>scripts/build_production_dc12_fixtures.py</code>	DC-001/DC-002 requirements define proxy rows and production schema fields.
<code>data/dc12_missing_production_telemetry.csv</code>	<code>data/production_dc12_telemetry_schema.csv</code>	Missing production fields become explicit required schema coverage.

Origin	Consuming artifact	Flow
scripts/ingest_production_dc12_telemetry.py	data/production_dc12_ingestion_results.csv	Fixture rows are validated, blocked, granted zero or positive candidate credit, and labeled production-calibrated or not.
data/production_dc12_ingestion_results.csv	data/production_dc12_claim_update_matrix.csv	Candidate, blocked, security, and control outcomes update claim status without upgrading production calibration.
tests/verify_production_dc12_telemetry.py	data/production_dc12_* and figures	Verifies schema coverage, fail-closed invalid rows, candidate-only valid rows, and nonblank figures.

The implementation boundary is now explicit: the architecture package can accept production evidence, but it has not yet observed production evidence that upgrades CL-012 or makes Option B/C production recommendations.